Government Documents

Everything a procurement officer, contracting specialist, or prime contractor needs. Print or download.

Capability Statement

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▶ The Cochran Block, LLC — Capability Statement

The Cochran Block, LLC
7452 School Avenue, Dundalk, MD 21222
mcochran@cochranblock.org · cochranblock.org
EIN: 41-3835237

Certifications & Registrations

SAM.gov — **Active** · CAGE 1CQ66 · UEI W7X3HAQL9CF9 · EIN 41-3835237
Maryland CSB (Certified Small Business) — **Approved**
SDVOSB (Service-Disabled Veteran-Owned Small Business) — Submitted (VetCert/SBA)
Maryland eMMA — Vendor SUP1095449 · ACH Direct Deposit Active
crates.io — 4 published crates (kova-engine, exopack, any-gpu, header-writer)
GitHub — 15+ repositoriesitories at github.com/cochranblock

Core Competencies

- Memory-safe compiled architecture (Rust) — aligned with CISA Secure-by-Design mandate
- Zero-cloud single-binary deployment — eliminates cloud attack surface entirely
- On-device AI inference — no data exfiltration, runs in air-gapped environments
- DevSecOps binary optimization — 48 KB to 10 MB release binaries with full SBOM
- Edge computing and IoT integration — LoRa/915MHz mesh, ARM/RISC-V targets
- Autonomous threat detection — 312 KB APT hunter, zero-config deployment

NAICS Codes

541511 — Custom Computer Programming Services
541512 — Computer Systems Design Services
541519 — Other Computer Related Services
518210 — Computing Infrastructure Providers
541330 — Engineering Services
541690 — Other Scientific and Technical Consulting

Past Performance

- **cochranblock.org** — Production website, 10 MB binary, $10/month, 16 products, intake forms, booking calendar, community grant app. Self-hosted on bare metal.
- **oakilydokily.com** — First paying client. Waiver management, digital intake, ESIGN compliance. Bare metal via Cloudflare Zero Trust.
- **Pixel Forge** — AI sprite generator with on-device diffusion models, 3 MoE models, LoRA fine-tuning. Pure Rust.

- **USCYBERCOM J38 JMOC-E** — Dev lead for Congressional NDAA-directed offensive cyber operations study.
- **15 public GitHub repos** — All code auditable at github.com/cochranblock

Differentiators

- **Proven federal contracting posture** — CAGE 1CQ66, UEI W7X3HAQL9CF9, SAM.gov Active
- **Single-binary = zero infrastructure** — 10MB replaces $36K/year cloud stacks for $120/year
- **Rust = memory-safe mandate compliance** — Aligned with CISA Secure-by-Design, EO 14028, NIST SP 800-218
- 4 published crates on crates.io: kova-engine, exopack, any-gpu, header-writer
- 15 Rust repositories with Proof of Artifacts and Timeline of Invention
- 4 inventions, 3 architecture patterns, 5 techniques — honestly classified at cochranblock.org/arch
- 13 years defense and enterprise — USCYBERCOM J38 dev lead, Congressional NDAA study
- Army 17C (Cyber Operations), JCAC 2014
- All Rights Reserved — The Cochran Block, LLC

Past Performance

- oakilydokily.com — First paying partnership. Waiver management, digital intake, ESIGN. Deployed on bare metal via Cloudflare Zero Trust.
- cochranblock.org — Live production site, 15 products, intake forms, SQLite, booking calendar, 10 MB binary, $10/month total infrastructure.
- USCYBERCOM J38 JMOC-E — Dev lead for Congressional NDAA-directed offensive cyber operations study.

Contact

Michael Cochran, Owner
mcochran@cochranblock.org
cochranblock.org/book — Schedule a call
cochranblock.org/deploy — Start a project

Downloadable Documents

▶ Available documents

W-9: Available on request —

Registration Status

Maryland eMMA	Vendor SUP1095449 — Active
Certified Small Business (CSB)	Approved
SAM.gov	Active · CAGE 1CQ66 · UEI W7X3HAQL9CF9
SDVOSB (VetCert)	Submitted — VetCert Apr 10, 2026
GSA Schedule	Not Yet Applied

Technical Approach — SBIR/STTR

▶ Zero-Cloud Edge Architecture for Defense and Federal Applications

Problem Statement

Federal agencies spend $36,000+/year per application on cloud infrastructure. These deployments create single points of failure, expose sensitive data to third-party providers, and require dedicated DevOps teams. Forward-deployed and disconnected environments cannot rely on cloud connectivity for mission-critical operations.

Technical Innovation

CochranBlock has developed a compiled single-binary architecture in Rust that eliminates cloud dependency entirely. A complete web application — server, database, authentication, TLS, asset pipeline — compiles into a 10 MB binary that runs on commodity hardware ($10/month total infrastructure). This architecture is proven in production at cochranblock.org, serving multiple domains from a single laptop.

Key Technical Capabilities

- **Single-binary deployment** — Entire application stack compiles to one executable. No containers, no orchestration, no package managers. Deploy by copying one file.
- **On-device AI inference** — Local LLM execution via custom Mixture-of-Experts architecture. No API calls, no data exfiltration risk. Runs on consumer GPUs.
- **Edge-native by design** — Operates in disconnected, intermittent, and limited-bandwidth (DIL) environments. Zero external dependencies at runtime.
- **Distributed C2 mesh** — Multi-node orchestration via SSH with tokenized command compression. Nodes operate independently and resync when connectivity restores.
- **Zero-trust architecture** — AES-256-GCM encryption, HKDF key derivation, Argon2id password hashing. No plaintext secrets in source. Cloudflare Zero Trust tunnel integration.
- **Embedded storage** — sled (embedded key-value store) + bincode serialization + zstd compression. No external database servers. Data lives with the application.
- **97% cost reduction** — Demonstrated: $36,000/year cloud → $120/year bare metal. Same availability, same performance, fraction of the attack surface.

Relevant SBIR/STTR Technology Areas

- AI/Autonomy — On-device inference, Mixture-of-Experts routing, edge AI for sensor processing
- Cybersecurity — Zero-trust compiled architecture, embedded encryption, no third-party attack surface
- Edge Computing — DIL-capable single-binary deployment, IoT gateway (LoRa/915MHz)
- Advanced Computing — Compiled Rust replacing interpreted cloud stacks, WASM-capable architecture
- Command and Control — Distributed node mesh with compressed tokenized command protocol

Proof of Concept — Live Production Systems

- **cochranblock.org** — 15-product portfolio site. 10 MB binary. SQLite intake forms. Booking calendar. Cloudflare tunnel. $10/month.

- **oakilydokily.com** — Waiver management and digital intake with ESIGN compliance. Bare metal deployment.
- **Kova augment engine** — Local LLM inference, agentic tool loop, distributed node C2, cargo/git tokenization. Single binary.
- **Approuter** — Reverse proxy with automatic Cloudflare tunnel management, app registry, multi-domain routing. Single binary.
- **15 repositoriesitories** — All source code publicly auditable at github.com/cochranblock

Phase I Objectives (6 months, $250K)

1. Formalize the single-binary deployment framework as a reusable platform for federal applications
2. Demonstrate on-device AI inference for classified/sensitive workloads with zero cloud dependency
3. Deploy proof-of-concept in a simulated DIL environment with multi-node mesh recovery
4. Deliver security assessment and ATO-ready documentation package

Phase II Path ($2M, 18–24 months)

1. Harden for IL4/IL5 deployment with FIPS 140-3 cryptographic module integration
2. Build agency-specific application templates (case management, intake, reporting)
3. Integrate with DoD identity providers (CAC/PIV authentication)
4. Develop training curriculum for agency adoption

Commercialization

The technology has immediate dual-use application. Commercial clients (startups, SMBs) pay $3,500 base deployment + $225/hour consulting. Federal clients access through SBIR Phase III, GSA Schedule 70, or direct contract. Year 1 projected gross: $150,000.

Principal Investigator

Michael Cochran — Army 17C (Cyber Operations), JCAC 2014. 13 years defense and enterprise. USCYBERCOM J38 JMOC-E dev lead for Congressional NDAA-directed offensive cyber operations study. 30% service-connected disabled veteran.

Agency-Specific Technical Approaches

▶ DoD SBIR 26.1 — Cyber-Resilient Edge Computing for Contested Environments

▶ NSF Seed Fund — On-Device AI Inference Without Cloud Dependency

- ▶ DHS/CISA — Zero-Trust Edge Architecture for Critical Infrastructure
- ▶ NASA — Edge Computing for Space and Aeronautics Ground Systems
- ▶ DOE — Cybersecurity for Energy Infrastructure and Scientific Computing
- ▶ USDA — Rural Broadband and Agricultural Edge Computing
- ▶ EPA — Environmental Monitoring for Air and Water Quality
- ▶ DOT — Edge Computing for Transportation Infrastructure
- ▶ NIST/Commerce — Software Supply Chain Security and SBOM
- ▶ NIH — Privacy-Preserving On-Device Health Data Processing
- ▶ NOAA — Remote Environmental and Ocean Monitoring

Upcoming Bids — SBIR/STTR 2026

Solicitation tracker. Updated as agencies publish topics. CochranBlock technical approach ready for each.

Agency	Solicitation	Opens	Closes	Status
DoD	SBIR 26.1 Phase I	April 2026	May 2026 (est)	SAM.gov Pending
DOE	Phase II Release 2	March 2, 2026	April 21, 2026	SAM.gov Pending
NIH	New NOFO	April 2026 (est)	Rolling	SAM.gov Pending
NSF	Seed Fund Restart	April–May 2026	Rolling	SAM.gov Pending
NASA	BAA Appendix A	April–May 2026	TBD	SAM.gov Pending
DHS/CISA	FY2026	Summer 2026	TBD	SAM.gov Pending

USDA	FY2026	June–Aug 2026	TBD	SAM.gov Pending
EPA	FY2026	Spring–Summer 2026	TBD	SAM.gov Pending
DOT	FY2026	Spring–Summer 2026	TBD	SAM.gov Pending
NIST	FY2026	Spring–Summer 2026	TBD	SAM.gov Pending
NOAA	FY2026	Spring–Summer 2026	TBD	SAM.gov Pending

SAM.gov Active. CAGE 1CQ66. UEI W7X3HAQL9CF9. SDVOSB submitted via VetCert.

Architecture & Compliance FAQ

▶ Q: Who owns the IP, and how do you handle Data Rights?

Zero vendor lock-in. All core code is All Rights Reserved. Government or Prime owns their deployment 100%. No recurring licensing. No proprietary runtime. No vendor dependency.

Every repo ships with a **Timeline of Invention** (TOI) and **Proof of Artifacts** (PoA) providing commit-level AI provenance documentation. This eliminates accidental copyright infringement risks from AI-generated code — every human decision and AI contribution is documented, dated, and hash-verified.

▶ Q: Who handles sustainment, patching, and DevOps?

No DevOps team required. We replace Kubernetes clusters and microservices with single, memory-safe, statically linked Rust binaries — often under 50 KB. No dependency chains to patch. No container images to rebuild. No orchestration layer to manage.

Patching = rebuild the binary from pinned Cargo.lock + SCP to the server. If the hardware has power, the binary runs. Total sustainment burden: one file.

▶ **Q: How does this survive DoD Authority to Operate (ATO) audits?**

Modern ATO delays are caused by bloated attack surfaces — cloud supply chains, Docker vulnerabilities, exposed Node/Python runtimes, 500+ transitive dependencies with unknown provenance.

Our architecture **eliminates that attack surface**:

- Zero cloud supply chain — no AWS, no Azure, no GCP dependency
- Zero unnecessary open ports — one binary, one port, behind Cloudflare Zero Trust
- Zero interpreted runtimes — compiled Rust, memory-safe by construction
- Complete SBOM at compile time — every dependency pinned and auditable
- NIST SP 800-218 (SSDF) compliance documented — see SSDF matrix below

Built by a former **USCYBERCOM J38 JMOC-E offensive cyber operations dev lead**. The architecture was designed to survive the audits, not to pass them after the fact.

▶ **Q: Is your operation DCAA compliant for SBIR Phase II/III or Prime subcontracts?**

Radical financial transparency. The Open Books page calculates IR&D hours and value in real-time from GitHub commit timestamps. Every hour is machine-verified, not self-reported. The methodology is public. The data is auditable.

IR&D costs documented per FAR 31.205-18. AI tooling costs separately trackable as materials under FAR 31.205-26. Founder hours valued at published $225/hr rate.

This is continuously auditable by design — not DCAA-compliant because we hired an accountant, but because the entire operation is transparent by architecture.

Software Bill of Materials (SBOM)

EO 14028 compliant. Every direct dependency, version, and license — known at compile time.

▶ **cochranblock — 42 direct dependencies**

NIST SP 800-218 — Secure Software Development Framework

▶ SSDF Compliance Matrix — cochranblock

CMMC Level 1-2 Practices

▶ CMMC Domain Mapping — cochranblock

Security Posture

▶ Cryptographic Primitives & Attack Surface

Cost Analysis

For a detailed cost comparison of cloud vs zero-cloud architecture: cochranblock.org/stats